

hypethesis

Palladium**■ What is it?**

Microsoft's ambitious plan to change the architecture of PCs in order to address concerns relating to security, privacy and intellectual property.

■ How does it work?

Microsoft is hoping to add these features to future versions of the PC with a combination of hardware and software. Basically, Palladium is about deciding what's trustworthy. It not only lets your computer know that you're you, but can also limit what runs on your computer. The system uses high-level encryption to lock data and can also protect the integrity of documents so that they can't be altered without your knowledge. Even the link between the keyboard and computer can be encrypted!

■ What does it mean to me?

Palladium is touted as the complete security and privacy solution. It tells you who you're interfacing with or who's using your PC and what they're doing, protects information from unauthorised access and modification, stops viruses and prevents spam. Your information can be transferred to other systems (say, a bank) by an 'agent'—even the agent can't read the information, it only carries it. It's currently being offered to record studios to control media piracy.

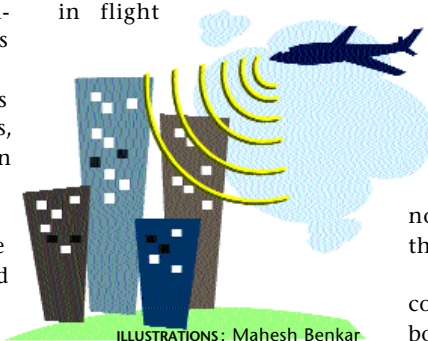
Solar plane for 3G

A California-based technology company, SkyTower, is planning to launch crewless solar-powered airplanes that will provide entire cities with broadband Internet access and 3G mobile services from 70,000 feet above the ground. SkyTower is working with NASA and the Japanese Ministry of Telecommunications on this project.

Tests show that SkyTower's planes, called Pathfinder-Plus, could complement and, in some cases, supersede satellite and terrestrial telecommunications systems. Because the signal is being transmitted from directly or almost directly above, the SkyTower sys-

tem will cover areas that cannot receive satellite or terrestrial signals due to obstructions such as large buildings.

The upper wing of the Pathfinder-Plus plane is made of solar panels, which generate the energy for the aircraft to remain in flight



ILLUSTRATIONS: Mahesh Benkar

throughout the day. The plane must land at night, but SkyTower is trying to create fuel cells that would store enough energy to allow the plane to stay circling a city overnight. Such fuel cells could enable an airplane to stay in the air for up to six months at a time. The extremely tight turning circle of the Pathfinder-Plus means that the plane stays within a small area throughout its flight, so people on the ground will not have to keep moving their receivers to track it.

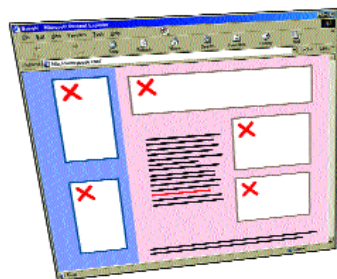
SkyTower is confident of a commercial launch of the airborne services in 2005.

Have JPEG, will pay

The ISO standards body will take the unprecedented step of withdrawing the JPEG image format as a formal standard if Forgent Networks, a small Texan company, continues to demand royalties on a very old patent. According to Richard Clark, JPEG committee member, under the ISO terms, only a standard that can be implemented for

free or on Reasonable and non-discriminatory (RAND) terms can be allowed. It is clear that Forgent's claims are not RAND compliant. "However, ISO itself cannot formally take a stance on the patent," Clark added. Unlike MPEG, there is no licensing structure for JPEG.

The current assumption is that the patent is not valid. Since the patent becomes invalid in 2004 (under the old rules, 17 years after filing the patent) there is little incentive for manufacturers to bow to the terms, though Forgent declined to rule out litigation against those who decline to pay them royalties on the claim.



A new XP

The upcoming version of Windows XP, named Windows XP Media Center Edition, aims to make the PC a permanent part of home entertainment.

It will enable users to use a TV remote control to catalogue songs, videos and pictures, as well as check TV listings. However, users will need a new system to run it on. Hewlett-Packard and Samsung are among the computer manufacturers developing Windows Media Center PCs and prices are estimated between \$1,000 and \$2,000.

